



Exp :3 Write the Python Program for Water Jug Problem

Aim:

To Write the Python Program for Water Jug Problem

Algorithm:

Step1: Start with two empty jugs of different capacities, and a target quantity of water to measure.

Step2: Fill one jug to its maximum capacity.

Step3: Pour the water from the full jug into the other jug until it is full or the first jug is empty.

Step4: If the second jug is full, pour out the water to empty it.

Step5: Repeat steps 2-4, filling and pouring water between the two jugs, until the desired quantity of water is measured or all possible combinations are tried.

Program:

```
from collections import deque
def water_jug_solver(capacity1, capacity2, target):
    visited = set()
    queue = deque([((0, 0), [])])
    while queue:
        (jug1, jug2), path = queue.popleft()
        if (jug1, jug2) in visited:
            continue
        visited.add((jug1, jug2))
        current_path = path + [(jug1, jug2)]

        if jug1 == target or jug2 == target:
            return current_path
```